Exercise Sheet 3

Generative Models and Naive Bayes

Probabilistic Machine Learning

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3.A Keep the Peak

Exercise 3.A.1: Peak, Mean and a Prior's Peak

Returning users, $\theta = P(\text{user returns})$, prior $\text{Beta}(a, b)$, h of n users return.

1. Maximise $\log p(D \mid \theta) + \log p(\theta)$ and show $\hat{\theta}_{\text{MAP}} = \frac{h+a-1}{n+a+b-2}$.
2. **0 of 4** users return. Give the MLE, the MAP under $\text{Beta}(2, 2)$ and the posterior mean under $\text{Beta}(2, 2)$. Which one would you report to a product manager, and why?
3. The posterior mean under a flat prior is $\frac{h+1}{n+2}$. Under which prior is the **MAP** equal to it?
4. After 3 of 3 with a flat prior the posterior is $\text{Beta}(4, 1)$, with density $4\theta^3$. Compute its central 95% interval.

Exercise 3.A.1: Peak, Mean and a Prior's Peak — Solution

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1. $\log p(D \mid \theta) + \log p(\theta) = (h + a - 1)\log\theta + (n - h + b - 1)\log(1 - \theta) + \text{const}$. Setting the derivative $\frac{h+a-1}{\theta} - \frac{n-h+b-1}{1-\theta}$ to zero gives the formula — the MLE of a dataset with $a - 1$ extra yeses and $b - 1$ extra noes.
2. MLE $0/4 = 0$: "nobody ever returns". MAP $\frac{0+1}{4+2} = \frac{1}{6} \approx 0.17$. Posterior $\text{Beta}(2, 6)$, mean $\frac{2}{8} = 0.25$. Report the posterior (mean and interval): four users are too few for a point, and the MLE claims certainty the data cannot give.
3. $\frac{h+a-1}{n+a+b-2} = \frac{h+1}{n+2}$ for all h, n needs $a = b = 2$: the MAP under $\text{Beta}(2, 2)$ equals the posterior mean under $\text{Beta}(1, 1)$. "Add one yes and one no" can be read either way — the Naive Bayes module of Lecture 03 calls it Laplace smoothing.
4. The CDF is θ^4, so the interval is $[0.025^{1/4}, 0.975^{1/4}] \approx [0.40, 0.99]$ — the peak at 1 lies at the edge of it.

3.B The Generative Route

Exercise 3.B.1: Counting Parameters

A fraud classifier (two classes) uses four attributes: **payment** (card, invoice, wallet), **country** (domestic, EU, other), **new device** (yes/no) and **night** (yes/no).

1. How many free numbers does the full generative model need — the class prior and one joint $p(x \mid y)$ per class?
2. How many does naive Bayes need?
3. For p yes/no attributes and K classes, give both counts as formulas.
4. From which p on does the full joint need more than a million numbers *per class*?

Exercise 3.B.1: Counting Parameters — Solution

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1. Full model. Each class needs a table over $3 \cdot 3 \cdot 2 \cdot 2 = 36$ value combinations, of which 35 are free (they sum to 1). Two classes give 70; the prior $P(\text{fraud})$ adds one: **71**.

2. Naive Bayes. Per class one categorical per attribute: $(3-1)+(3-1)+(2-1)+(2-1) = 6$ free numbers. Two classes give 12, plus the prior: **13**.

3. Formulas. Full: $K(2^p - 1) + (K - 1)$. Naive Bayes: $Kp + (K - 1)$.

4. $2^{19} - 1 = 524{,}287$ and $2^{20} - 1 = 1{,}048{,}575$, so from $p = 20$ on. The naive count grows linearly: 20 per class.

Exercise 3.B.2: Fitting Is Counting

An attribute takes k values. Within one class we observed it n times, value v exactly n_v times ($\sum_v n_v = n$). Model it as categorical with parameters $\theta_1, \ldots, \theta_k \geq 0, \sum_v \theta_v = 1$.

1. Write down the log-likelihood $\ell(\theta)$ of the n i.i.d. observations.
2. For $k = 2$, substitute $\theta_2 = 1 - \theta_1$ and show that the MLE is $\hat{\theta}_1 = n_1/n$ — the h/n of Lecture 03.
3. For general k, maximise ℓ under the constraint $\sum_v \theta_v = 1$ (Lagrange multiplier) and show $\hat{\theta}_v = n_v/n$.
4. Why can naive Bayes fit every attribute's table separately, while a full joint $p(x \mid y)$ could not be fitted attribute by attribute?

Exercise 3.B.2: Fitting Is Counting — Solution

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1. Each observation of value v contributes θ_v:

$$\ell(\theta) = \sum_{v=1}^{k} n_v \log \theta_v$$

2. $\ell = n_1 \log \theta_1 + n_2 \log(1-\theta_1)$. Setting $\ell'(\theta_1) = n_1/\theta_1 - n_2/(1-\theta_1) = 0$ gives $n_1(1-\theta_1) = n_2\theta_1$, so $\hat{\theta}_1 = n_1/(n_1+n_2) = n_1/n$.

3. $L(\theta, \lambda) = \sum_v n_v \log \theta_v - \lambda(\sum_v \theta_v - 1)$. From $\partial L/\partial\theta_v = n_v/\theta_v - \lambda = 0$ we get $\theta_v = n_v/\lambda$; the constraint gives $\lambda = \sum_v n_v = n$, so $\hat{\theta}_v = n_v/n$ — the relative frequency.

4. Under the naive assumption the log-likelihood of the table is a **sum** of one term $\sum_v n(x_j = v, y) \log \theta_{jvy}$ per attribute and class; no parameter appears in two terms, so each term is maximised on its own — by counting. A full joint has one parameter per *combination* of values; its counts are counts of whole rows, and they cannot be split into per-attribute counts.

3.C Naive Bayes by Hand

Exercise 3.C.1: Twelve Transactions

Flags: new device D, foreign IP F, night N, high amount H (1 = yes).

#	D	F	N	H	fraud?	#	D	F	N	H	fraud?
1	1	1	1	1	yes	7	1	0	0	0	no
2	1	1	0	1	yes	8	0	0	0	0	no
3	1	0	1	0	yes	9	0	1	0	0	no
4	0	1	1	1	yes	10	0	0	0	0	no
5	0	0	0	0	no	11	1	0	1	0	no
6	0	0	1	0	no	12	0	0	0	0	no

1. Estimate $P(\text{fraud})$ and $P(\text{flag} = 1 \mid \text{class})$ for all four flags by counting.
2. Compute $P(\text{fraud} \mid x)$ for $x_1 = (D, F, N, H) = (1, 0, 0, 0)$.
3. Do the same for $x_2 = (0, 0, 0, 1)$. What goes wrong, and why?
4. Repeat 2 and 3 with Laplace smoothing on the flags (keep the prior unsmoothed). Which result changes a lot, which only a little?

Exercise 3.C.1: Twelve Transactions — Solution

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1. $P(\text{fraud}) = 4/12 = 1/3$. Among the 4 frauds, each flag occurs 3 times: $3/4$ for D, F, N, H. Among the 8 honest: $D\ 2/8, F\ 1/8, N\ 2/8, H\ 0/8$.

2. Fraud: $\frac{1}{3} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{256} \approx 0.0039$. Honest: $\frac{2}{3} \cdot \frac{2}{8} \cdot \frac{7}{8} \cdot \frac{6}{8} \cdot 1 = 0.1094$. $P(\text{fraud} \mid x_1) = 0.0039/(0.0039 + 0.1094) = 1/29 \approx 0.034$.

3. Fraud: $\frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{1}{256}$. Honest: $\ldots \cdot P(H{=}1 \mid \text{honest}) = 0$. So $P(\text{fraud} \mid x_2) = 1$: a transaction without any red flag except a high amount is **certain** fraud, because no honest high amount occurred in eight rows. One zero count vetoes the other three flags, which all speak for honest.

4. Smoothed, $(\text{count} + 1)/(n + 2)$: fraud $4/6 = 2/3$ for every flag; honest $D\ 0.3, F\ 0.2, N\ 0.3, H\ 0.1$. x_1: fraud $\frac{1}{3} \cdot \frac{2}{3} \cdot (\frac{1}{3})^3 = 0.0082$, honest $\frac{2}{3} \cdot 0.3 \cdot 0.8 \cdot 0.7 \cdot 0.9 = 0.1008$, so $P = 0.075$ (from 0.034). x_2: fraud 0.0082, honest $\frac{2}{3} \cdot 0.7 \cdot 0.8 \cdot 0.7 \cdot 0.1 = 0.0261$, so $P \approx 0.24$ (from 1). With 4 and 8 rows the prior of one pseudo-count per value is not small, so even x_1 moves — smoothing matters most where the counts are smallest.

Exercise 3.C.2: Smoothing Is a Prior

A yes/no flag was seen h times among n frauds.

1. With a $\mathrm{Beta}(1,1)$ prior on $\theta = P(\mathrm{flag} \mid \mathrm{fraud})$, what is the posterior (Lecture 02)? Show that its **mean** is Laplace's $(h+1)/(n+2)$.
2. What is the posterior's **peak** for $h = 0$? What does that say about "keep the peak"?
3. "Add-α" smoothing uses $(h+\alpha)/(n+2\alpha)$. Which prior belongs to it? Compute the estimate for $h = 0, n = 100$ and $\alpha = 0.5, 1, 10$.
4. A merchant category with $k = 10$ values; one category never occurred in 100 frauds. What does Laplace smoothing give, and which prior is behind it?

Exercise 3.C.2: Smoothing Is a Prior — Solution

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1. Adding counts: $\mathrm{Beta}(1+h,\ 1+n-h)$, with mean $\frac{1+h}{(1+h)+(1+n-h)} = \frac{h+1}{n+2}$.

2. The mode of $\mathrm{Beta}(a,b)$ is $\frac{a-1}{a+b-2}$; here $\frac{h}{n} = 0$. Under a flat prior the peak *is* the MLE, so keeping the peak keeps the zero. Laplace smoothing silently switches to the mean.

3. $\mathrm{Beta}(\alpha,\alpha)$: posterior $\mathrm{Beta}(\alpha+h, \alpha+n-h)$, mean $(h+\alpha)/(n+2\alpha)$. For $h=0, n=100$: $\alpha = 0.5$: $0.5/101 \approx 0.0050$; $\alpha = 1$: $1/102 \approx 0.0098$; $\alpha = 10$: $10/120 \approx 0.083$. A larger α is a stronger belief that θ is near $1/2$ — worth 2α imagined observations.

4. $(0+1)/(100+10) = 1/110 \approx 0.0091$ — the posterior mean under a flat **Dirichlet** $(1,\ldots,1)$, the Beta for more than two values.